

Friction: Friend and Enemy

Week 3 · Lesson A · Monday · 40 minutes

Name: _____ Date: _____

I can explain what friction does and give examples where it helps and where it hinders.

Start with what you notice

Hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds. Stop if it feels uncomfortable. You can watch the adult or describe a familiar example instead.

Choose your four-question arrival page: Supported 7, Standard 8 or Stretch 9. Use screen or paper; you do not need to do both.

Watch the adult test two surfaces

The adult uses the same block, ramp angle and release point. Only the surface changes. First, predict. Then record only what you actually see.

Before the test	My prediction
Surface I think will resist sliding more	
My reason	

We changed: _____. We kept the same: _____.

What I observed (or label it “model only” if there was no real test):

A short run is evidence of greater resistance in this controlled setup. It is not a direct force measurement. Smooth-looking surfaces can still grip.

Choose the job; judge the force

Discuss, point, draw, write or dictate. An adult can scribe your words.

Helpful or unhelpful?

Start with examples 1 and 3. For each, identify the two touching surfaces and the job. Use the other examples if there is time.

Situation	Helpful, unhelpful or need context?	Which contact / why?
1. Bike brakes		
2. Trainers on a wet floor		
3. Engine parts rubbing		
4. Pushing a heavy box across carpet		
5. A door handle		

Sort the claims

Choose TRUE, FALSE or NOT A SCIENTIFIC CLAIM. Explain a choice aloud or on your selected response page. The cut-out version is on page 6.

Claim	My decision
A. Sliding friction opposes sliding at the contact.	
B. Friction only happens with rough surfaces.	
C. Oil can reduce sliding friction in suitable lubricated contacts.	
D. Without friction between our shoes and the ground, we could not walk in the usual way.	
E. Friction is a bad force.	

Keep the decision tied to the contact and the job. No oil is needed for this activity.

CHOOSE THIS PAGE / SUPPORTED

Show one helpful example

Use with common pages 1–2. You do not need the other two response pages.

Word bank: friction · surfaces · grip · rubbing · slowing · heat · wear

Point, say, draw or write

Friction acts where two _____ touch and slide, or try to slide.

Choose one: a bike brake / a hand on a door handle. Draw or name the two touching surfaces.

Friction is helpful here because _____

Now choose an unhelpful example

Choose one: engine parts rubbing / pushing a box across carpet.

Friction is unhelpful for this job because _____

Check one claim

Can smooth-looking surfaces still grip? Say, point or write one example.

STOP after the shared claim sort. Put your name on your work and keep it for Tuesday. Your prediction comes with you.

CHOOSE THIS PAGE / STANDARD

Explain both sides

Use with common pages 1–2. You do not need the other two response pages.

Draw the force

Draw a box sliding to the right across a floor. Add a labelled arrow for the sliding friction acting on the box.

Build a comparison

Use the two touching surfaces in each explanation.

Friction is helpful on _____ because _____.

But it is unhelpful for _____ because _____.

Correct a claim

“Friction only happens with rough surfaces.” Explain why this is false. Use the smooth rubber / smooth glass example discussed in class.

STOP after the shared claim sort. Label and keep your work for Tuesday. Keep your surface prediction; do not claim a test you did not perform.

CHOOSE THIS PAGE / STRETCH

Make the explanation precise

Use with common pages 1–2. You do not need the other two response pages.

Same force, different job

Compare bike brakes with engine parts. Identify the contacts, explain the effect of friction and explain why the effect helps or hinders the intended job. Include energy transfer or wear.

Test the wording

Someone says: “A smooth surface always has less friction.” Use the smooth rubber / smooth glass example to challenge this. What evidence would you need to compare two surfaces fairly?

A changed case

Two unknown materials touch. Is their friction helpful? Explain what extra information you need before deciding.

STOP after the shared claim sort. Label and keep your work for Tuesday. Your next lesson tests surfaces; today’s reasoning is not a completed investigation.

REUSABLE CARDS / OPTIONAL PRINT

Cut, place and explain

Use instead of writing in the tables. An adult can cut the cards before the lesson.

Helpful / unhelpful: sort by the job. Start with bike brakes and engine parts.

Card	Situation
1	Bike brakes
2	Trainers on a wet floor
3	Engine parts rubbing
4	Pushing a heavy box across carpet
5	A door handle

Claim cards

Card	Claim
A	Sliding friction opposes sliding at the contact.
B	Friction only happens with rough surfaces.
C	Oil can reduce sliding friction in suitable lubricated contacts.
D	Without friction between our shoes and the ground, we could not walk in the usual way.
E	Friction is a bad force.

Sorting labels

TRUE	FALSE	NOT A SCIENTIFIC CLAIM

Place cards in three groups below these labels, or point to a label. Give a reason; moving a card alone is not the whole explanation.

ARRIVAL / SUPPORTED

Friction arrival

Supported · Four minutes within Lesson A

Choose one route. Point, say, sign, draw or write. You may pass or say not sure. Use screen or paper; you do not need to do both.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

1. A push moves a toy car away. Which action moved it? Push / pull.	2. Look at two palms pressed together. Which surfaces touch? Palm and palm / palm and sleeve.
3. Use your chosen route. What do you notice when hands rub? Say, point or draw one observation.	4. A shoe needs to grip the ground. Does grip help walking or make slipping easier? Help walking / make slipping easier.

Choose one response route. An adult can read or scribe your words.

ARRIVAL / STANDARD

Friction arrival

Standard · Four minutes within Lesson A

Choose one route. Point, say, sign, draw or write. You may pass or say not sure. Use screen or paper; you do not need to do both.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

1. Name the two actions that can change how an object moves: p___ and p___.	2. When hands rub, name the two touching surfaces.
3. Use your chosen route. Describe one change you notice or already know. If you know the force's name, add it.	4. A bike brake pad presses on the wheel rim. Predict what happens to the turning wheel.

Choose one response route. An adult can read or scribe your words.

ARRIVAL / STRETCH

Friction arrival

Stretch · Four minutes within Lesson A

Choose one route. Point, say, sign, draw or write. You may pass or say not sure. Use screen or paper; you do not need to do both.

For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.

1. A toy car is pushed, then slows on a level floor. Name one force you think could slow it. Not sure is useful.	2. A hand grips a door handle. Name the touching surfaces and explain what the grip helps you do.
3. Use your chosen route. Suggest why rubbing hands may feel warmer. Give a prediction if you cannot observe it.	4. Two toy cars start at different speeds on different surfaces. Is that a fair surface comparison? Name one thing to keep the same.

Choose one response route. An adult can read or scribe your words.

KEEP NEARBY / KNOWLEDGE ORGANISER

Friction knowledge organiser

Learning objective: Explain what friction does and give examples where it helps and where it hinders.

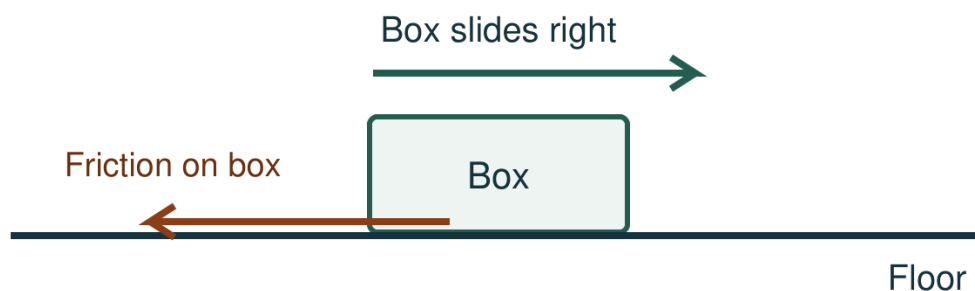
Success: Name the two touching surfaces. Explain how friction helps or hinders the intended job. Use an observation or a clearly labelled model to support an idea.

Word	Meaning
Friction	A force that resists sliding, or attempted sliding, between touching surfaces.
Surface	The outside face of an object where contact happens.
Grip	Friction that helps prevent slipping.
Wear	Damage caused by repeated rubbing.
Lubricant	A substance that can reduce friction in suitable contacts.

Friction acts at a contact. Identify both surfaces and their relative sliding.

The actual surface pair and its condition matter. Smooth rubber and smooth glass can still grip; roughness alone does not predict every result.

Rubbing can transfer energy to thermal stores. A brief rub need not produce noticeable warmth.



For a box sliding right on a floor, friction on the box acts left along the contact. This is an explanatory model, not a force measurement.

Helpful: pad-wheel friction helps a bike brake slow the wheel.

Unhelpful: rubbing between moving engine parts can cause unwanted heating and wear.

Use evidence: Compare surfaces using the same block, fixed ramp and release point, no push, and the same measuring start. Repeat a run. Record what actually happened; label supplied results as model data.

CHOOSE ONE ROUTE / TWO COPIES TO CUT

Friction exit Supported

Lesson A · Supported

Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

1. A box slides right on a floor. Which way does friction on the box act?

Left / Right

2. Bike brake pads rub on the wheel rim. Does this help slow the wheel?

Yes / No

Next lesson: Friction: test the surfaces

Lesson A · Supported

Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

1. A box slides right on a floor. Which way does friction on the box act?

Left / Right

2. Bike brake pads rub on the wheel rim. Does this help slow the wheel?

Yes / No

Next lesson: Friction: test the surfaces

CHOOSE ONE ROUTE / TWO COPIES TO CUT

Friction exit Standard

Lesson A · Standard

Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

1. Friction acts between touching ____ and resists their sliding.

2. Friction at the brake pad helps the wheel ____.

Next lesson: Friction: test the surfaces

Lesson A · Standard

Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

1. Friction acts between touching ____ and resists their sliding.

2. Friction at the brake pad helps the wheel ____.

Next lesson: Friction: test the surfaces

CHOOSE ONE ROUTE / TWO COPIES TO CUT

Friction exit Stretch

Lesson A · Stretch

Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

1. Friction at a bike brake is helpful because ____.

2. Friction between moving engine parts can be unhelpful because ____.

Next lesson: Friction: test the surfaces

Lesson A · Stretch

Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.

1. Friction at a bike brake is helpful because ____.

2. Friction between moving engine parts can be unhelpful because ____.

Next lesson: Friction: test the surfaces